

A Gold/Silver Hedge for a Dynamic SPY/TLT Frontier

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Abstract

This research note tests whether a liquid precious-metals sleeve improves a dynamic SPY/TLT allocation. The core sleeve holds SPY/TLT at 60/40, the metals sleeve holds GLD/SLV at 70/30, and the neutral reference assigns 80

Disclaimer. This document is research software output, not investment advice, and not evidence of future returns.

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1 Thesis and research design

This note formalises an investment thesis as a testable mathematical object and evaluates it on historical data. The pipeline is deliberately mechanical: a strategy maps the price history available at time t to target weights (Section 2); a backtest converts those weights into a net-of-cost return series; and every number quoted in the text is generated from that series, not typed by hand (Sections 3–5 define the estimators; Section 10 reports them).

The research sample contains 930 sessions from 2023-01-03 to 2026-09-18. Trading costs are charged at 5 basis points per unit of turnover. These assumptions are fixed before the results are read, and they determine what the figures can and cannot show.

The key design rule is *no look-ahead*: the weight vector applied to the return of session t may depend only on information dated strictly before that return is realised. Formally, if $\mathcal{F}_{t-1}$ is the information set at the close of session $t - 1$, then

$$w_t = g(\mathcal{F}_{t-1}), \quad r_t^{\text{strat}} = \sum_i w_{i,t} r_{i,t} - c \sum_i |w_{i,t} - w_{i,t}^{\text{drift}}|, \quad (1)$$

where $w_{i,t}^{\text{drift}}$ is the previous target weight after drifting with the market and $c = \text{CostBps}/10^4$ is the proportional cost rate.

1.1 Hypotheses

- H1 The sinusoidal rule has a higher net Sharpe ratio than the static 80/20 core/metals reference.
- H2 The static metals-hedged reference has a smaller maximum drawdown than unhedged static 60/40 SPY/TLT.
- H3 The ranking of sine, linear, and quadratic interpolation changes across transaction-cost assumptions.

2 Portfolio accounting

2.1 Currency conversion

Let asset i be quoted in currency Q_i and let the account base currency be B . If X_t^{B,Q_i} denotes units of Q_i per one unit of B , a native price $P_{i,t}^{Q_i}$ has base-currency value

$$P_{i,t}^B = \frac{P_{i,t}^{Q_i}}{X_t^{B,Q_i}}. \quad (2)$$

Equivalently, with a direct multiplier $Y_t^{Q_i,B} = \left(X_t^{B,Q_i}\right)^{-1}$ in units of B per unit of Q_i ,

$$P_{i,t}^B = P_{i,t}^{Q_i} Y_t^{Q_i,B}. \quad (3)$$

Forward-filling an FX series onto an asset's trading dates is an alignment convention, not an estimate of an unobserved intraday rate. When all assets share the base currency, both equations are the identity.

2.2 NAV and target weights

With cash C_t , position quantity $q_{i,t}$, and base-currency price $P_{i,t}^B$, net asset value (NAV) is

$$V_t = C_t + \sum_{i=1}^m q_{i,t} P_{i,t}^B. \quad (4)$$

For target weights $w_i \geq 0$ satisfying $\sum_i w_i \leq 1$, the ideal target quantity is

$$q_{i,t}^* = \frac{w_i V_t}{P_{i,t}^B}. \quad (5)$$

Equation (5) is the bridge between economic allocation and position size. Unallocated weight, $1 - \sum_i w_i$, remains in cash and earns zero. The backtest uses fractional quantities so that rounding does not obscure the mathematics.

3 Returns, wealth, and performance

For a consistently adjusted price series P_t , the one-period simple return is

$$r_t = \frac{P_t}{P_{t-1}} - 1, \quad (6)$$

and the log return is

$$\ell_t = \log\left(\frac{P_t}{P_{t-1}}\right) = \log(1 + r_t). \quad (7)$$

Simple returns aggregate across simultaneously held assets; log returns add through time.

Starting from wealth V_0 , compounded wealth after T returns is

$$V_T = V_0 \prod_{t=1}^T (1 + r_t). \quad (8)$$

Total return and compound annual growth rate over Y calendar years are

$$R_{\text{tot}} = \frac{V_T}{V_0} - 1, \quad \text{CAGR} = (1 + R_{\text{tot}})^{1/Y} - 1. \quad (9)$$

Using $A = 252$ sessions per year and a zero risk-free rate, annualised volatility and the Sharpe ratio are

$$\hat{\sigma}_{\text{ann}} = \hat{\sigma}_r \sqrt{A}, \quad \hat{S} = \frac{\bar{r}}{\hat{\sigma}_r} \sqrt{A}. \quad (10)$$

For the rolling graph, $\bar{r}$ and $\hat{\sigma}_r$ are recomputed over a trailing window using the sample standard deviation; the estimate is path-dependent and moves sharply when an extreme observation enters or leaves the window.

Let $M_t = \max_{0 \leq s \leq t} V_s$ be the running high-water mark. Drawdown and maximum drawdown are

$$D_t = \frac{V_t}{M_t} - 1, \quad D_{\text{max}} = \min_{0 \leq t \leq T} D_t. \quad (11)$$

These summaries describe one realised path. They do not quantify parameter uncertainty, multiple testing, tail severity beyond the observed sample, or the probability of future loss.

4 Estimation and uncertainty

4.1 Gaussian maximum likelihood

For returns $x_1, \dots, x_n$, assume as a working approximation that $x_j \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$. The log-likelihood, up to an additive constant, is

$$\mathcal{L}(\mu, \sigma^2) = -\frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{j=1}^n (x_j - \mu)^2. \quad (12)$$

Maximisation yields

$$\hat{\mu} = \frac{1}{n} \sum_{j=1}^n x_j, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{j=1}^n (x_j - \hat{\mu})^2. \quad (13)$$

The divisor n is the maximum-likelihood convention, not the unbiased one. The estimated standard error of the mean is

$$\widehat{\text{se}}(\hat{\mu}) = \frac{\hat{\sigma}}{\sqrt{n}}. \quad (14)$$

For the daily strategy returns of this study, $\hat{\mu} = 6.238e-04$, $\hat{\sigma}^2 = 5.772e-05$ and $\widehat{\text{se}}(\hat{\mu}) = 2.491e-04$. An estimated edge is only interesting relative to this error: the statistic $\hat{\mu}/\widehat{\text{se}}(\hat{\mu})$ is a t -type ratio, not a significance test, because returns are neither established as Gaussian nor as independent.

4.2 Stationary bootstrap for the Sharpe ratio

Daily returns are serially dependent enough that the iid interval is unreliable. The stationary bootstrap of Politis and Romano resamples blocks with geometric random length. With mean block length $1/p$, define the index sequence recursively: I_1 is uniform on $\{1, \dots, n\}$, and

$$I_{k+1} = \begin{cases} (I_k \bmod n) + 1, & \text{with probability } 1 - p, \\ \text{uniform on } \{1, \dots, n\}, & \text{with probability } p. \end{cases} \quad (15)$$

For replication $b = 1, \dots, B$ (here $B = 1000$), form $x_k^{*b} = x_{I_k}$ for $k = 1, \dots, n$ and compute $\hat{S}^{*b}$ from Equation (10). The percentile interval for the Sharpe ratio is

$$\left[\hat{S}_{(\alpha/2)}^*, \hat{S}_{(1-\alpha/2)}^* \right], \quad (16)$$

the empirical $\alpha/2$ and $1 - \alpha/2$ quantiles of the replications. For this study the interval is $[0.31, 2.34]$ around the point estimate 1.30. The interval captures sampling noise under the resampling assumptions only; it does not correct for choosing the strategy after seeing data.

5 Position sizing

For a small risky return R and fraction f of wealth invested, a second-order expansion of expected log growth is

$$\mathbb{E}[\log(1 + fR)] \approx f \mathbb{E}[R] - \frac{f^2}{2} \mathbb{E}[R^2]. \quad (17)$$

When the mean is small relative to volatility, $\mathbb{E}[R^2] \approx \text{Var}(R)$, and maximising over f gives the familiar approximate optimiser

$$f^* \approx \frac{\mathbb{E}[R]}{\text{Var}(R)}. \quad (18)$$

Because $\mathbb{E}[R]$ and $\text{Var}(R)$ are estimated with error (Section 4), the full-Kelly fraction is systematically too aggressive. A fractional-Kelly rule scales it by $\lambda \in (0, 1]$ and imposes a cap $f_{\max}$, after subtracting the per-trade cost c :

$$f = \min\left(f_{\max}, \lambda \frac{\max(\hat{\mu} - c, 0)}{\hat{\sigma}^2}\right). \quad (19)$$

This is “Kelly-style”, not an exact Kelly solution: return distributions are not established, parameters are estimated, and the resulting weights are then constrained to $\sum_i w_i \leq 1$ as in Section 2. Growth is concave in f around f^* , so a fraction $\lambda < 1$ gives up a small amount of expected growth in exchange for a substantially smaller variance and drawdown.

For this study, plugging the estimates of Section 4 into Equation (18) gives an approximate full-Kelly fraction of 10.81; with $\lambda = 0.50$ and cap $f_{\max} = 1.00$, Equation (19) gives 1.00. A fraction above one would imply leverage; the strategy does not use these figures, which serve only to show how fragile the full-Kelly value is given the standard error above.

6 Thesis-specific model

GLD seeks to reflect gold-bullion prices less trust expenses, while SLV seeks to track silver bullion. They are liquid securities and imperfect commodity proxies, not direct delivery contracts. To control estimation error, the four assets are reduced to two internally fixed sleeves. Their returns are

$$c_t = 0.60r_{\text{SPY},t} + 0.40r_{\text{TLT},t}, \quad m_t = 0.70r_{\text{GLD},t} + 0.30r_{\text{SLV},t}. \quad (20)$$

For a metals allocation $h \in [0.05, 0.35]$, the four-asset weight vector is

$$w(h) = (0.60(1 - h), 0.40(1 - h), 0.70h, 0.30h)^\top. \quad (21)$$

The static hedged reference has $h_0 = 0.20$, hence weights $(0.48, 0.32, 0.14, 0.06)^\top$.

Let $x_t = (c_t, m_t)^\top$. Using at most $L = 126$ completed sessions, the rolling estimates are

$$\hat{\mu}_t = 252 \bar{x}_t, \quad \hat{\Sigma}_t = 252 \widehat{\text{Cov}}_t(x) + \epsilon I. \quad (22)$$

The global minimum-variance metals share and the mean-variance growth anchor are

$$h_t^{\text{gmV}} = \arg \min_h v(h)^\top \hat{\Sigma}_t v(h), \quad h_t^* = \arg \max_{h \in \mathcal{F}_t} \left\{ v(h)^\top \hat{\mu}_t - \frac{\gamma}{2} v(h)^\top \hat{\Sigma}_t v(h) \right\}, \quad v(h) = (1 - h, h)^\top, \quad (23)$$

where $\gamma = 3$ and $\mathcal{F}_t$ is the estimated efficient segment from the GMV point to the higher-estimated-return endpoint.

The five-session compound-growth gap between the anchor and static hedged reference is

$$d_t = \prod_{j=0}^4 \left(1 + w(h_t^*)^\top r_{t-j} \right) - \prod_{j=0}^4 \left(1 + w(h_0)^\top r_{t-j} \right). \quad (24)$$

If $\hat{\sigma}_{5,t}$ and $\hat{\sigma}_{21,t}$ are annualised volatilities of the static hedged reference, the raw coordinate is

$$q_t = \text{clip}\left(\frac{\hat{\sigma}_{5,t}/\hat{\sigma}_{21,t} - 1}{0.50}, -1, 1\right) \min\left\{\frac{|d_t|}{\hat{\sigma}_{21,t}\sqrt{5/252}}, 1\right\}. \quad (25)$$

Positive q_t moves toward the lower-risk point; negative q_t moves toward the frontier’s higher-risk endpoint. A zero growth gap leaves the allocation at the rolling anchor.

For interpolation type k , the causal five-session coordinate is

$$\tilde{q}_t^{(k)} = \sum_{j=1}^5 \kappa_j^{(k)} q_{t-5+j}, \quad \kappa_j^{(k)} = \frac{g_k(j)}{\sum_{\ell=1}^5 g_k(\ell)}, \quad (26)$$

with $g_{\text{sine}}(j) = \sin(\pi j/10)$, $g_{\text{linear}}(j) = j$, and $g_{\text{poly}}(j) = j^2$. The target formed after close t is applied to the return on session $t + 1$.

7 Empirical comparison

Table 1 separates two questions: whether adding a static metals sleeve helped relative to unhedged 60/40, and whether dynamically timing that sleeve added value.

Table 1: Net performance from January 2023.

Rule	CAGR	Volatility	Sharpe	Max drawdown	Avg. turnover
Sine	16.17%	12.07%	1.30	-11.85%	3.91%
Linear	16.16%	12.06%	1.30	-11.87%	3.93%
Quadratic	16.04%	12.04%	1.30	-12.12%	4.04%
Static hedged	15.92%	11.24%	1.37	-10.81%	—
Static 60/40	12.32%	11.19%	1.09	-12.72%	—

The pre-registered outcomes are H1 **Not supported**, H2 **Supported**, and H3 **Supported**. These labels describe one historical sample, not a population result.

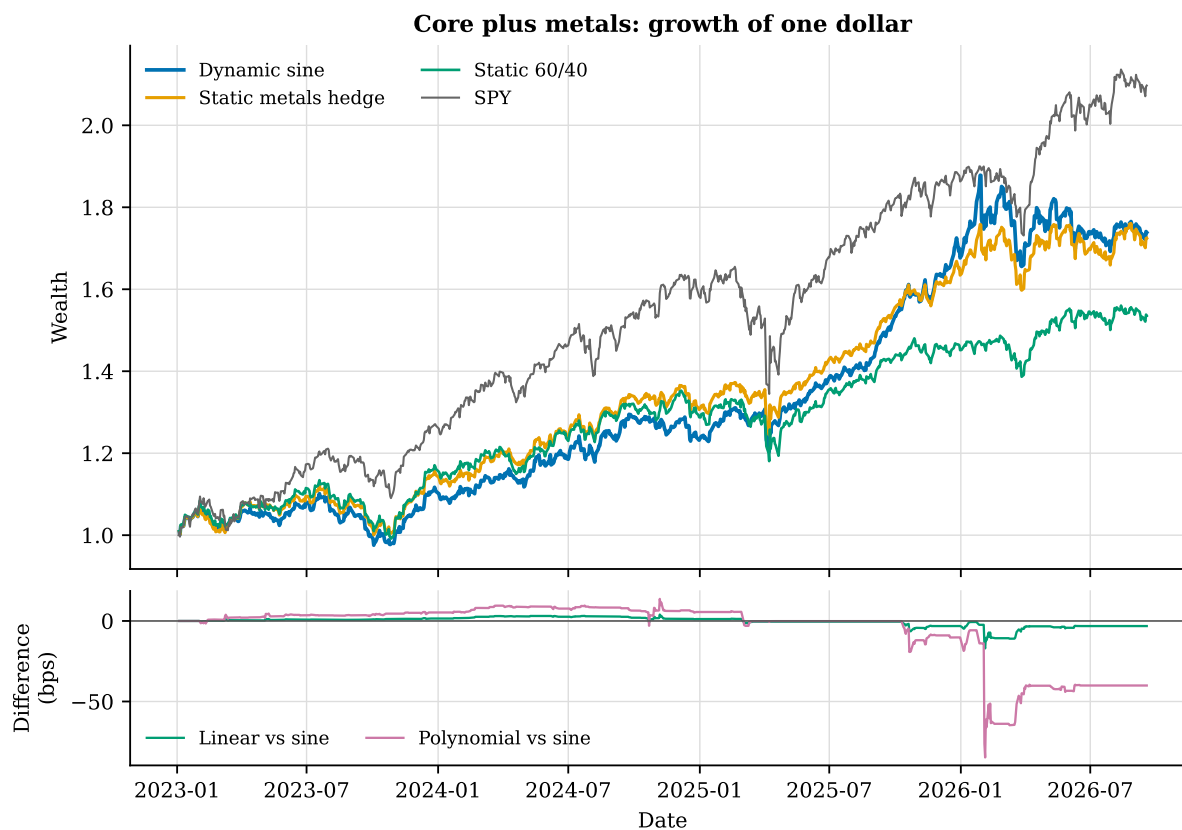


Figure 1: Top: cumulative wealth for the primary sine rule, static references, and SPY. Bottom: linear and quadratic cumulative-wealth differences from sine, in basis points.

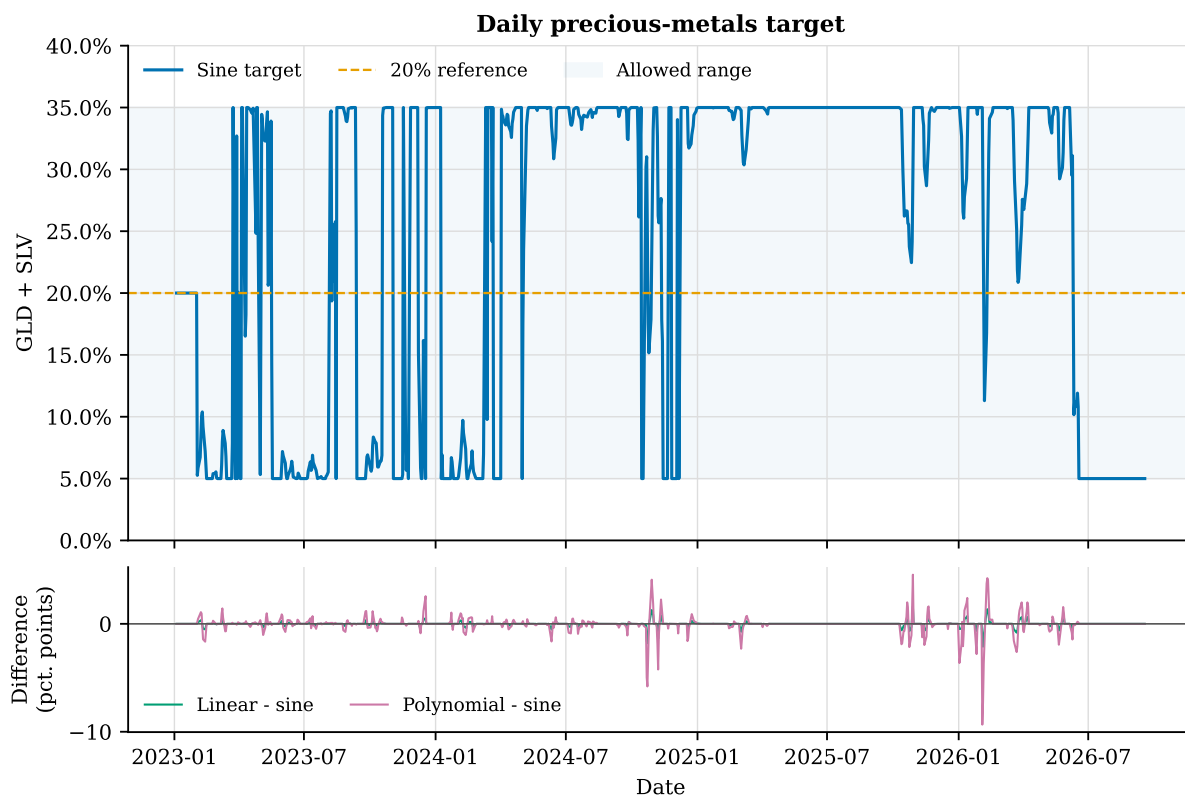


Figure 2: Top: sine-rule metals target and the static 20% reference. Bottom: target-weight differences between the alternative kernels and sine.

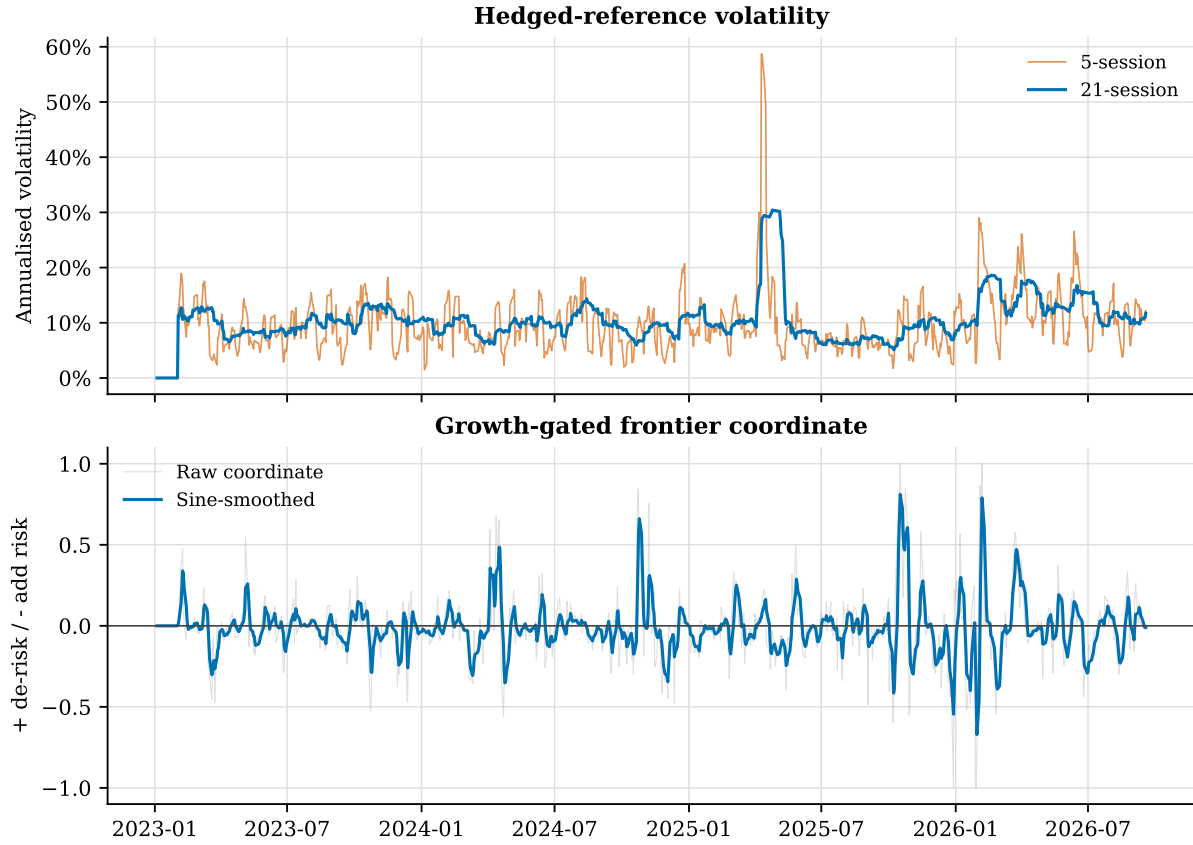


Figure 3: Reference volatility estimates and the sine-smoothed frontier coordinate; the noisy raw coordinate is shown faintly for context.

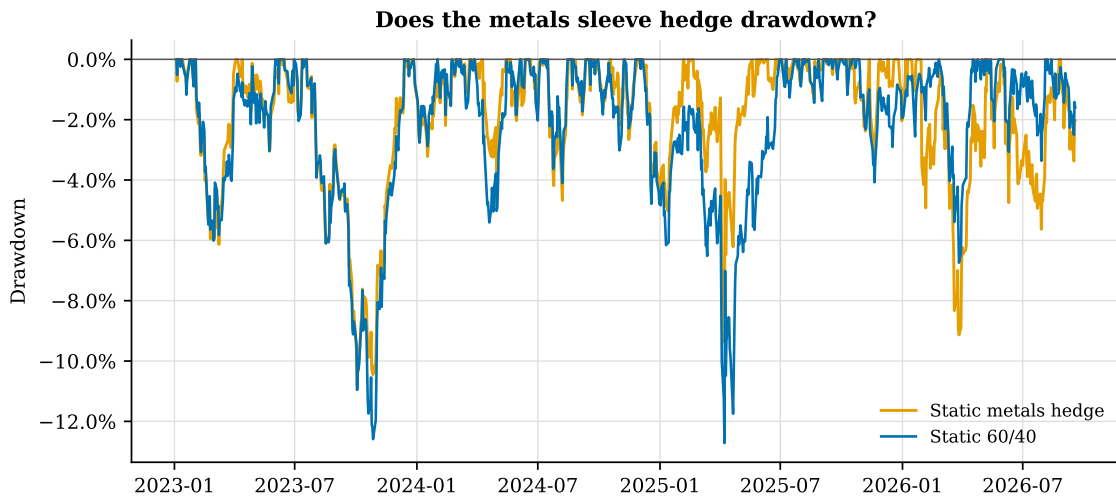


Figure 4: Drawdown paths for the static metals-hedged reference and unhedged 60/40.

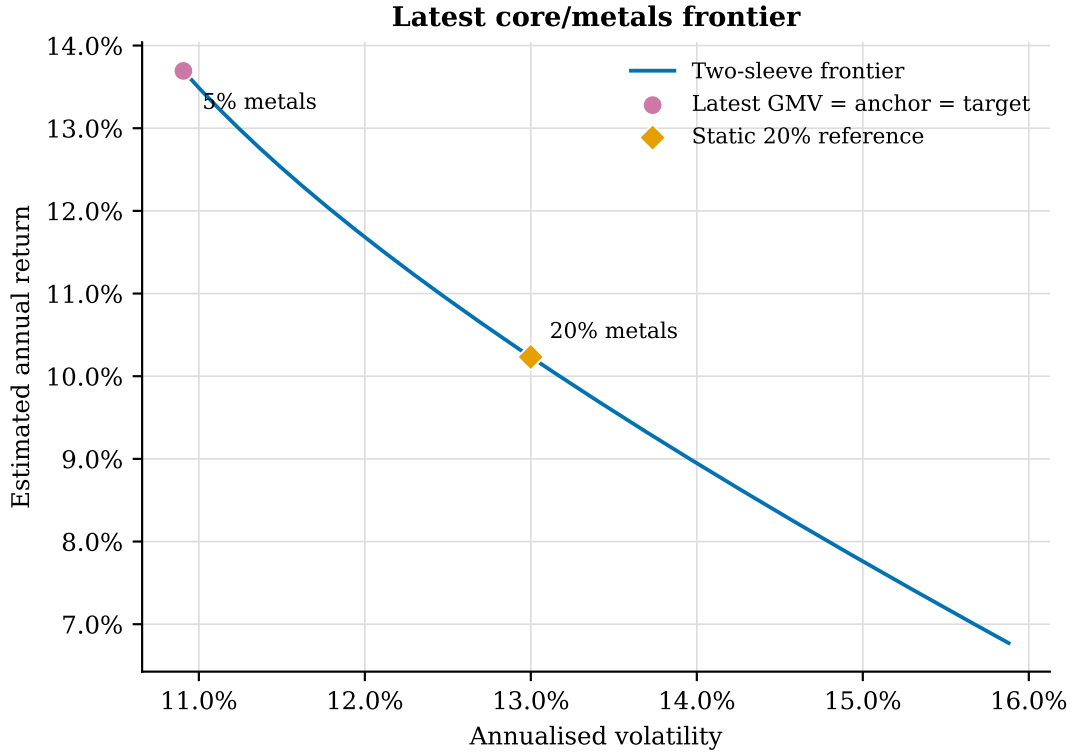


Figure 5: Latest estimated frontier between the fixed core and metals sleeves. Coincident GMV, anchor, and target points are collapsed into one labelled marker.

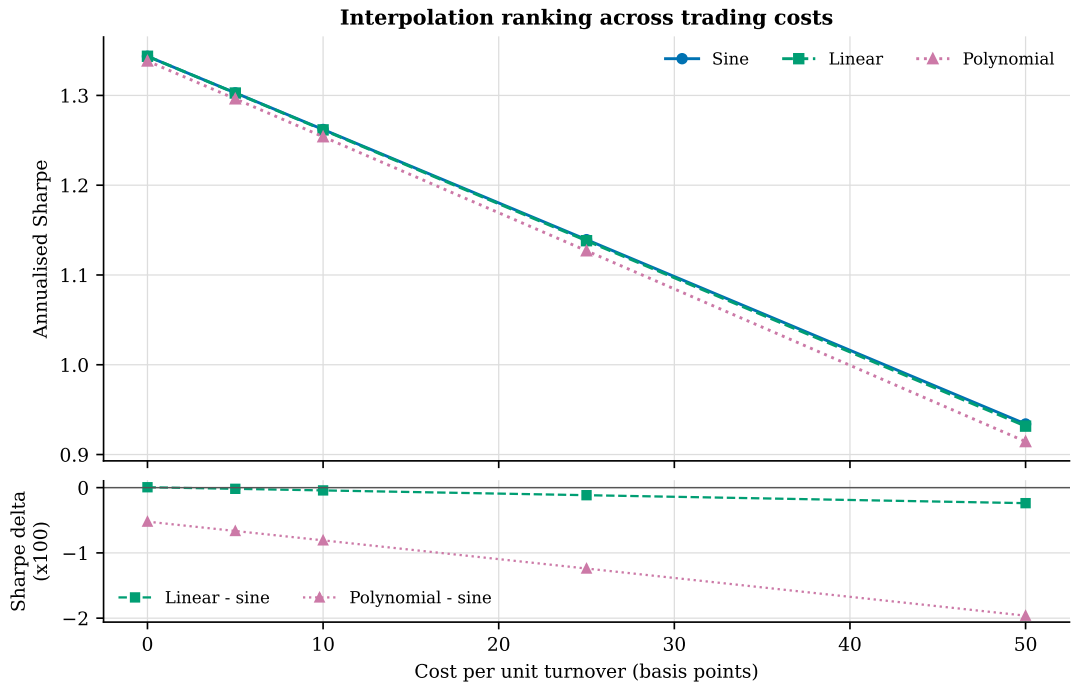


Figure 6: Top: interpolation Sharpe ratios under alternative trading costs. Bottom: linear and quadratic differences from sine, magnified by 100.

8 Trade and broker interpretation

The latest completed-session target, dated 2026-09-18, is 57.00% SPY, 38.00% TLT, 3.50% GLD, and 1.50% SLV. Its smoothed coordinate is 0.00. These are model weights, not an executable recommendation. The supplied IBKR planner checks for paper mode, reads current positions and base-currency prices, sizes whole-share differences, and sets `transmit=False`; it never places an order.

9 References

- H. Markowitz (1952), “Portfolio Selection,” *Journal of Finance* 7(1), 77–91, doi:10.1111/j.1540-6261.1952.tb01525.x.
- World Gold Trust Services, “SPDR Gold Shares: GLD,” official fund page.
- iShares, “iShares Silver Trust: SLV,” official fund page.
- D. N. Politis and J. P. Romano (1994), “The Stationary Bootstrap,” *JASA* 89(428), 1303–1313, doi:10.1080/01621459.1994.10476870.

10 Results

All values below are generated from the backtest output; none is typed by hand. The sample spans 930 sessions from 2023-01-03 to 2026-09-18 with costs of 5 basis points.

Table 2: Performance summary (net of costs). The interval for the strategy Sharpe ratio is the stationary-bootstrap percentile interval of Equation (16).

	Strategy	Benchmark
CAGR	16.17%	22.22%
Sharpe ratio	1.30	1.42
Maximum drawdown	-11.85%	-18.76%
Total return	73.87%	
Annualised volatility	12.07%	
Sharpe 95% interval	[0.31, 2.34]	

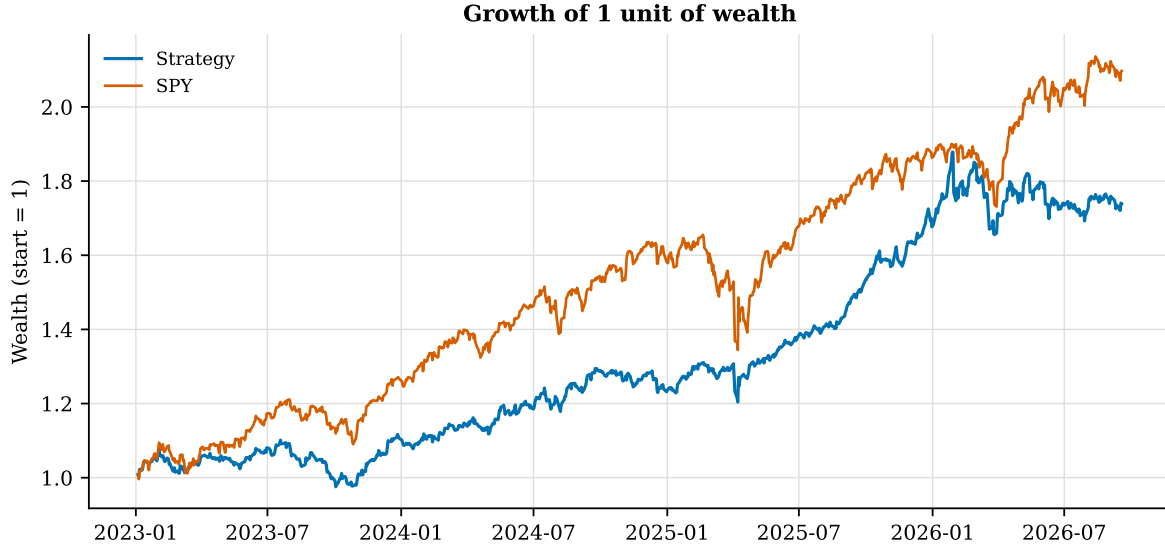


Figure 7: Cumulative wealth of the strategy and its benchmark (Equation (8)).

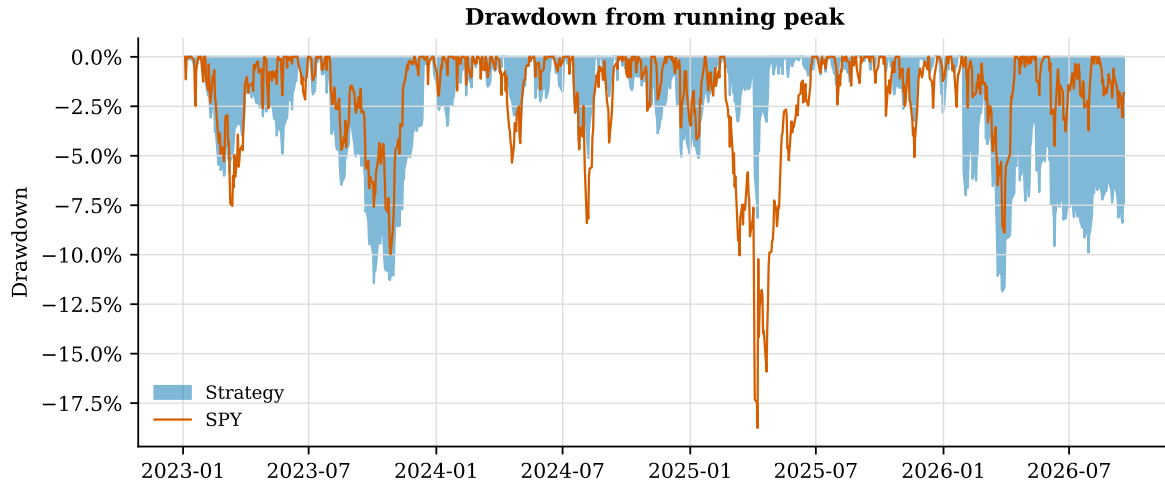


Figure 8: Drawdown from the running peak (Equation (11)).

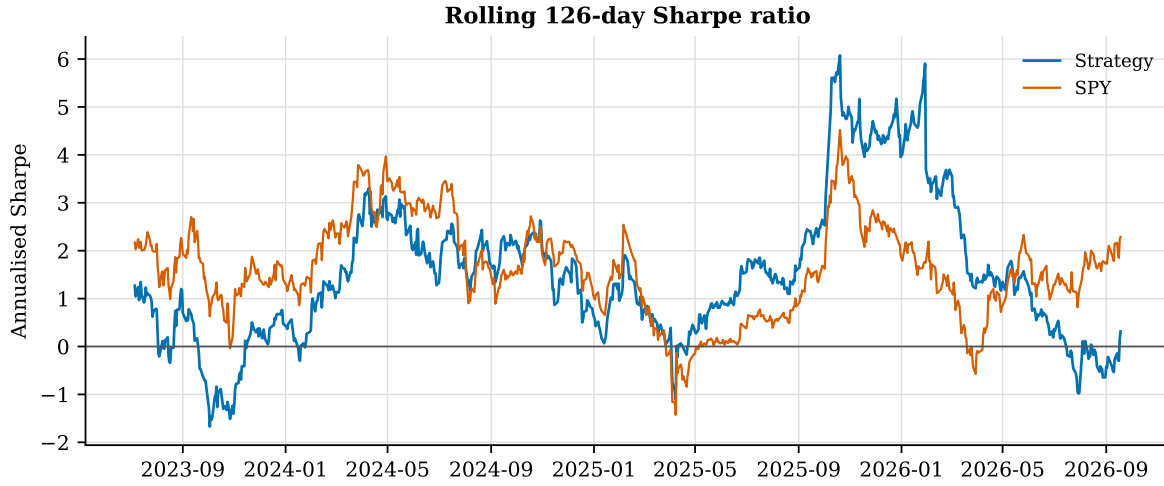


Figure 9: Rolling annualised Sharpe ratio (Equation (10)). Adjacent windows overlap almost completely, so the path is descriptive rather than a sample of independent estimates.

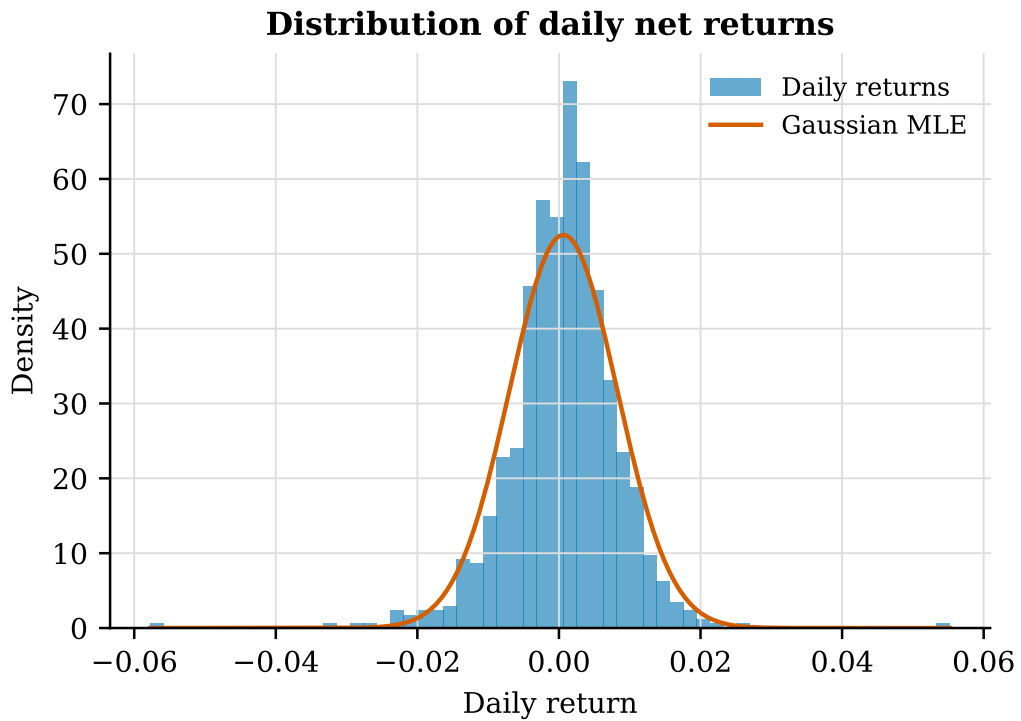


Figure 10: Distribution of daily strategy returns. Fat tails and skew, if visible, contradict the Gaussian working assumption of Equation (12).

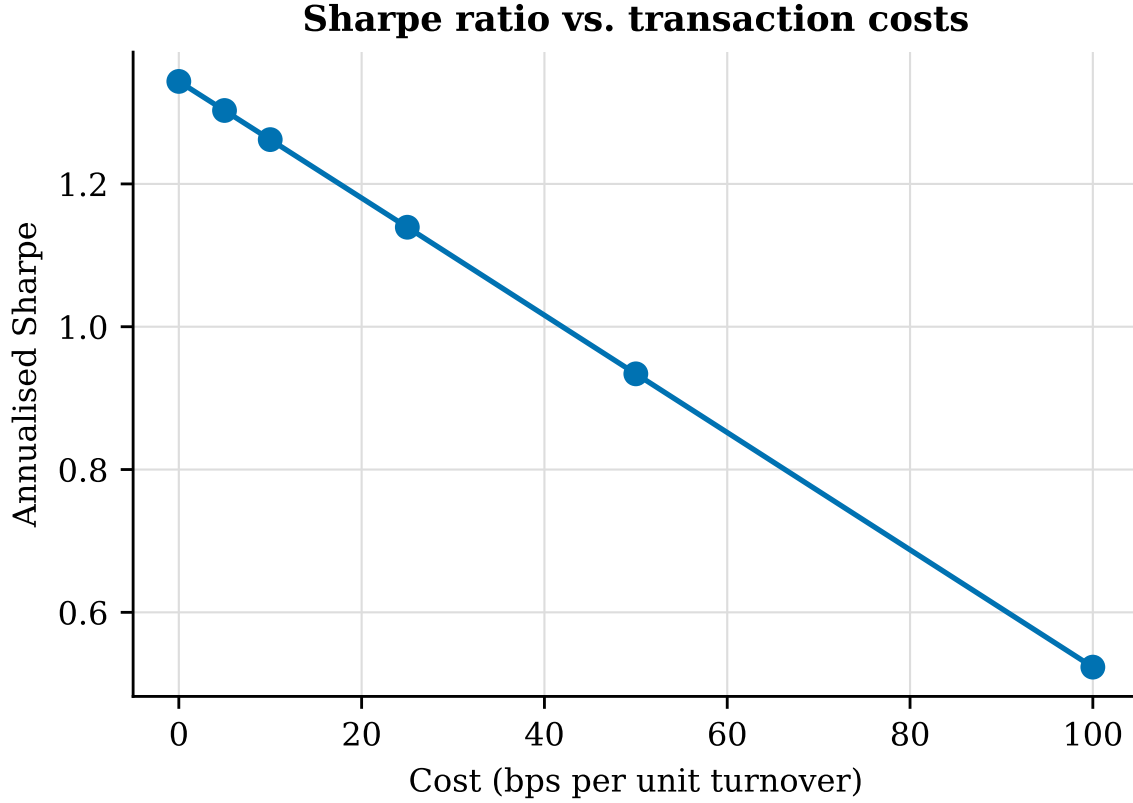


Figure 11: Sensitivity of performance to the transaction-cost assumption. A result that only survives at low cost is fragile.

11 Evidence and limitations

The figures show one realised path and are not evidence of dominance. The following caveats apply to every paper produced by this pipeline, whatever the numbers say.

- **Sample length.** The sample has 930 sessions (2023-01-03 to 2026-09-18). The standard error of a Sharpe ratio estimated from T years of data is roughly $\sqrt{(1 + S^2/2)/T}$, so short samples cannot distinguish skill from luck; see the interval [0.31, 2.34].
- **Costs.** Costs are a proportional 5 basis points per unit of turnover. Real costs include spreads that widen in stress, market impact, commissions, financing, and taxes, none of which are modelled.
- **Overfitting and multiple testing.** The thesis was formulated with knowledge of history. Every parameter choice, ticker choice, and discarded variant increases the chance that the best-looking result is noise. The bootstrap interval does not account for this selection.
- **Model assumptions.** Gaussian and iid assumptions are working approximations. Financial returns are heteroskedastic, skewed, and heavy-tailed.
- **Data.** Adjusted closes are assumed correct and free of survivorship bias; the risk-free rate is taken as zero, and rankings can change with other choices.

- **No out-of-sample test.** Unless stated otherwise, no untouched holdout period was reserved.

This document is not investment advice. It is an automatically generated research note; nothing in it has been validated for live capital, and past performance says little about future returns.

Conclusion

The chain of argument is mechanical: consistent prices give returns (Equation (6)); weights that use only past information give a net-of-cost strategy return (Equation (1)); compounding gives wealth, growth, and drawdown (Equations (8)–(11)); and estimation theory gives an honest statement of uncertainty (Equation (16)). Over 2023-01-03 to 2026-09-18 the strategy delivered a CAGR of 16.17% and a Sharpe ratio of 1.30 against 22.22% and 1.42 for the benchmark, with a maximum drawdown of -11.85%. Given the limitations of Section 11, these are hypotheses for further out-of-sample study, not conclusions about the future.